



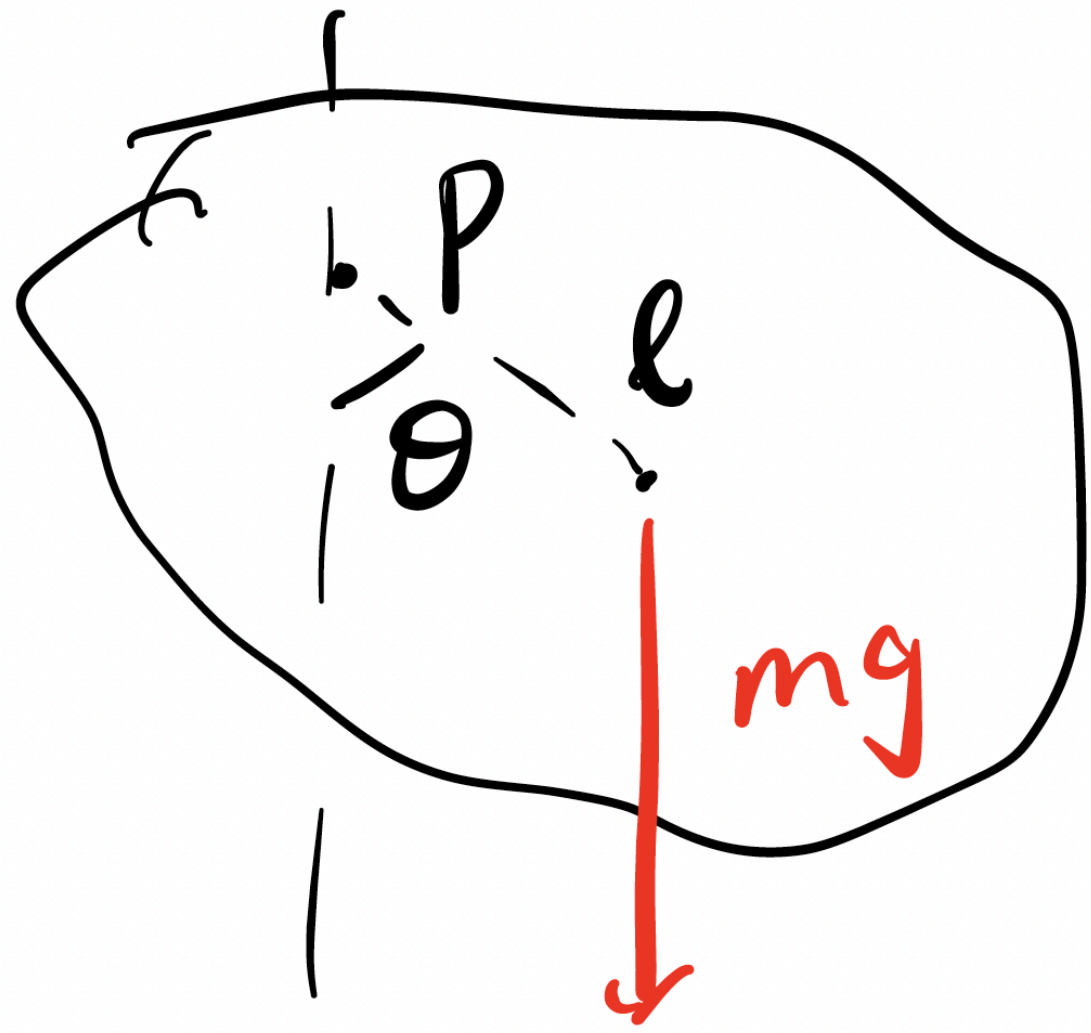
香港中文大學(深圳)
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PHY1001 Mechanics

Lecture 24: Oscillation

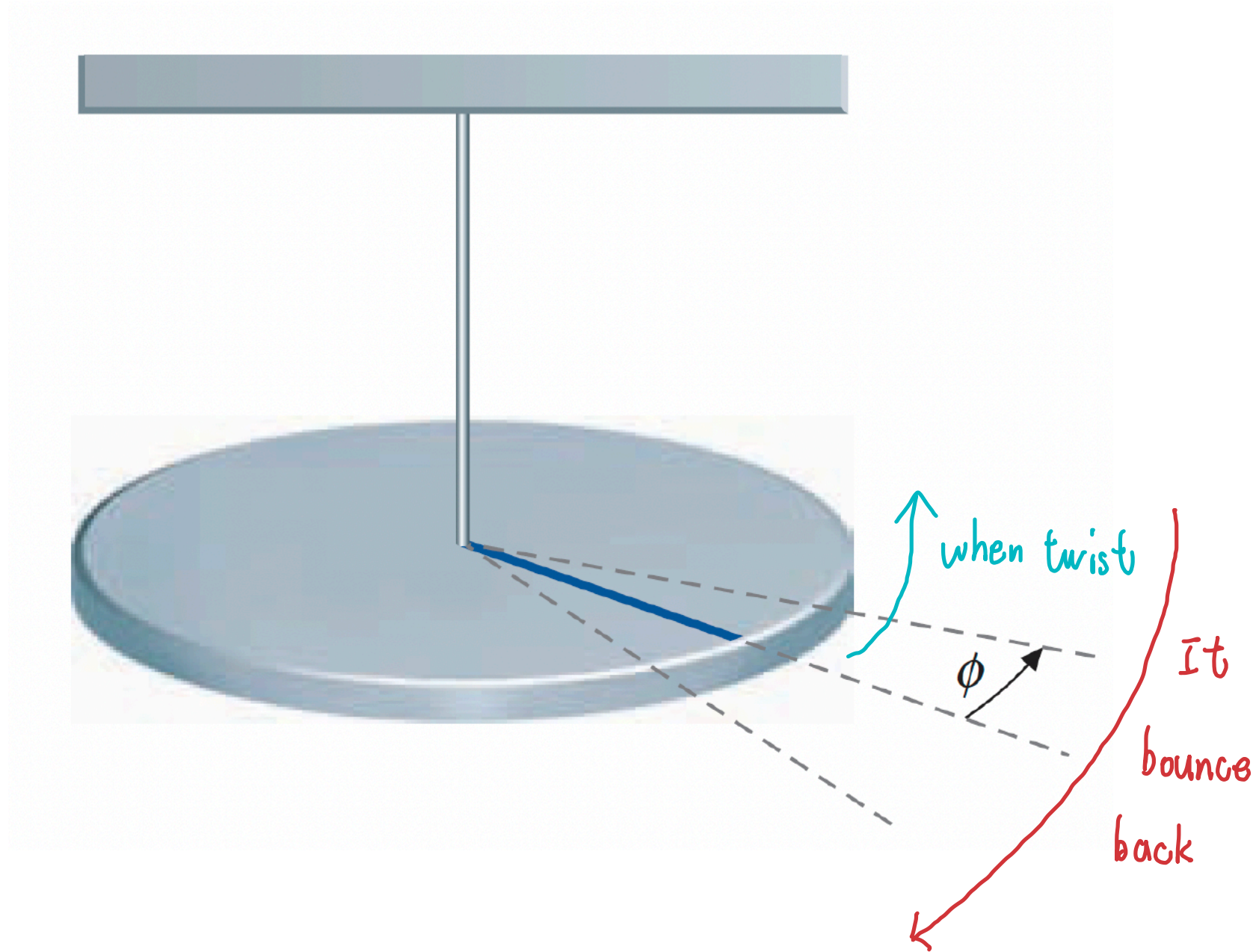
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Recap: Oscillation



- Physical pendulum
 - $\frac{d^2\theta}{dt^2} + \frac{mgl}{I}\theta = 0$
 - $\omega = \sqrt{\frac{mgl}{I}}$

Torsional Pendulum



$$\tau = -k\phi \quad \text{restoring torque}$$

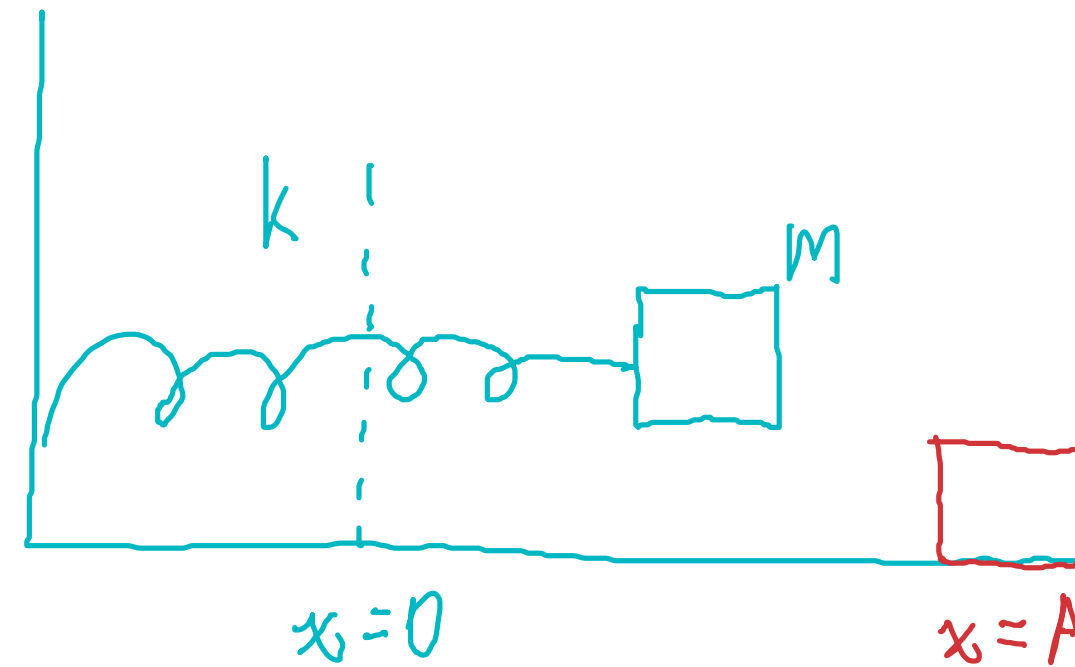
$$\tau = I\alpha$$

$$-k\phi = I \frac{d^2\theta}{dt^2}$$

$$I \frac{d^2\theta}{dt^2} + k\phi = 0$$

Energy of Oscillation

- $E = \frac{1}{2}kA^2$



$$x = A \cos(\omega t + \varphi)$$

$$E = K + U$$

$$= \frac{1}{2} k A^2$$

$$k = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2$$

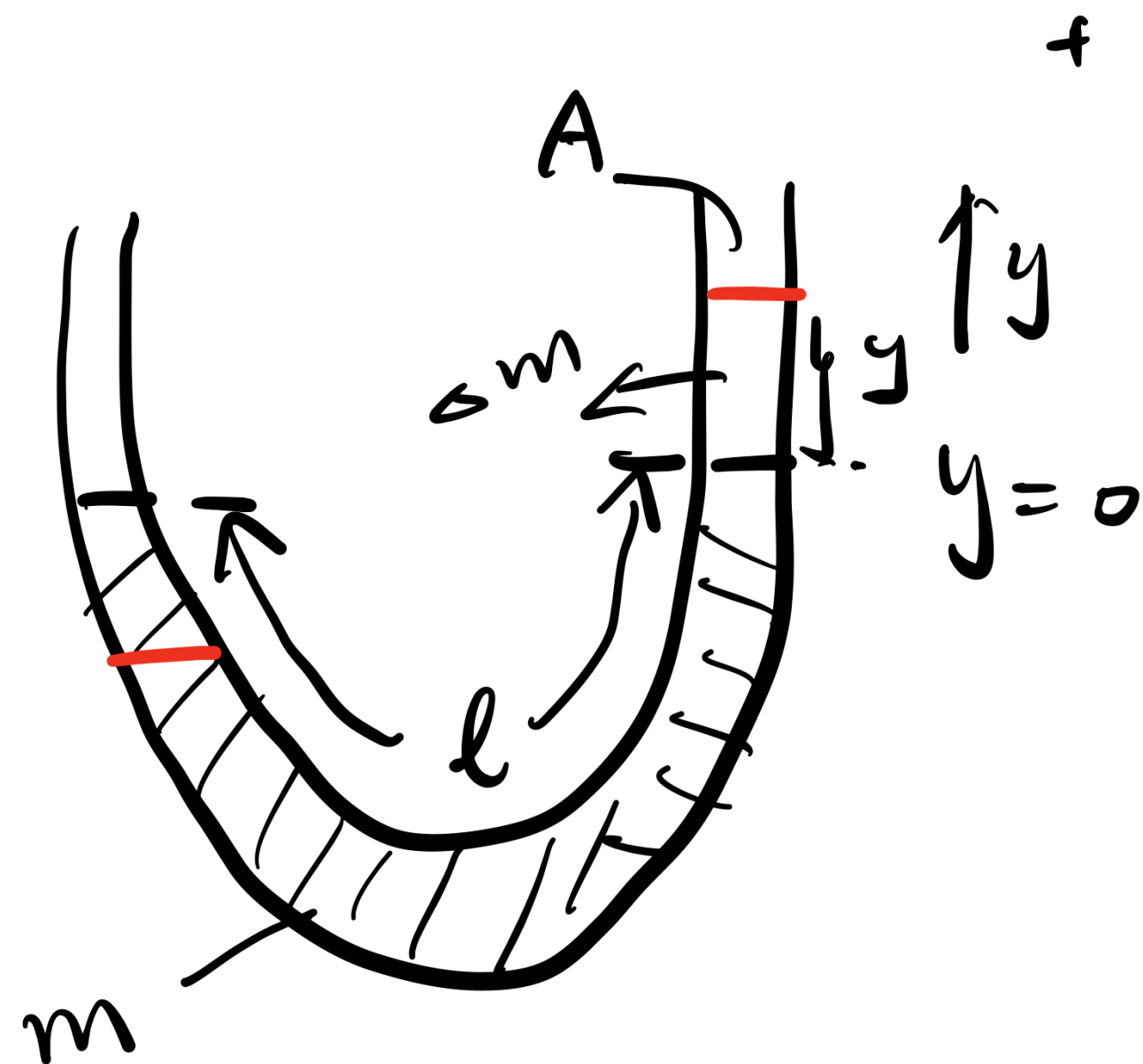
$$= \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \varphi) = \frac{1}{2} m \cdot \frac{k}{m} A^2 \sin^2(\omega t + \varphi)$$

$$U = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k A^2 \cos^2(\omega t + \varphi)$$

$$\omega^2 = \frac{k}{m}$$

Fluid & Oscillation Revist



$$K + U_g = \text{const.}$$

$$U_g = \Delta m \cdot g \cdot y$$

$$= \rho \cdot A \cdot y \cdot g \cdot y = \rho \cdot A \cdot g \cdot y^2$$

$$K = \frac{1}{2} m v^2 = \frac{1}{2} \rho A l \cdot \left(\frac{dy}{dt} \right)^2$$

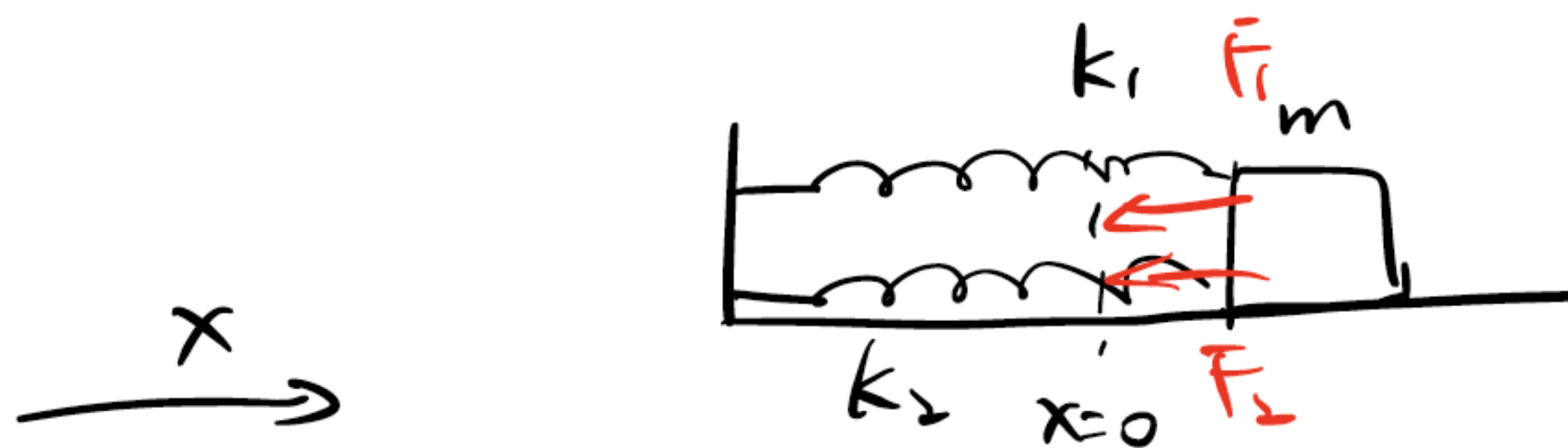
$$\frac{dE}{dt} = 0$$

$$\frac{1}{2} \rho A l \cdot 2 \frac{dy}{dt} \cdot \frac{d^2 y}{dt^2} + 2 \rho A g \cdot y^2 \cdot \frac{dy}{dt} = 0$$

$$l \cdot \frac{d^2 y}{dt^2} + 2gy = 0$$

$$\omega = \sqrt{\frac{2g}{l}}$$

Springs in Parallel



find ω

$$F_1 = -k_1 x$$

$$F_2 = -k_2 x$$

$$F = -(k_1 + k_2)x$$

effective spring constant

$$F = ma = m \frac{d^2 x}{dt^2} = -(k_1 + k_2)x$$

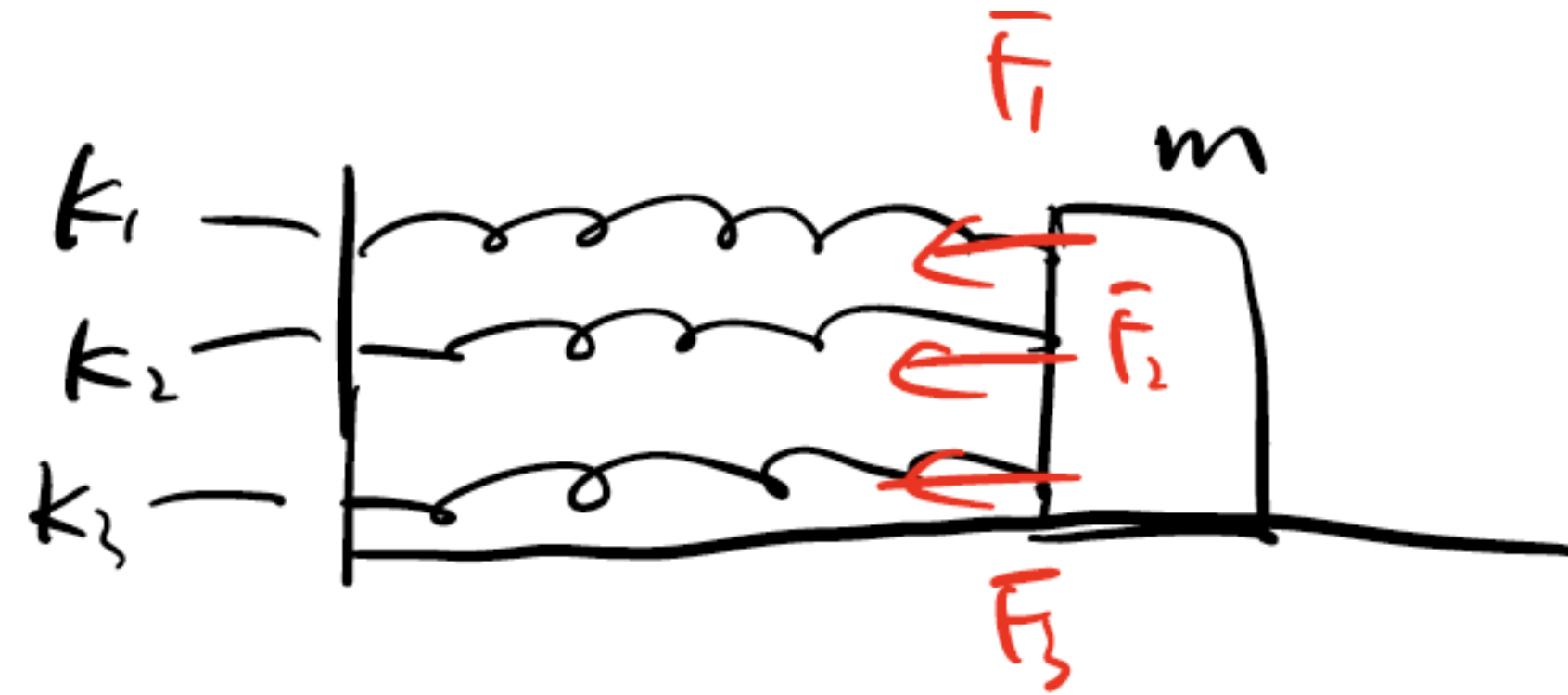
$$\frac{d^2 x}{dt^2} + \frac{k_1 + k_2}{m} x = 0$$

$$\omega = \sqrt{\frac{k_1 + k_2}{m}}$$

$$k = k_1 + k_2$$

$$\omega^2 = \frac{k}{m} = \frac{k_1 + k_2}{m}$$

Springs in Parallel



n springs

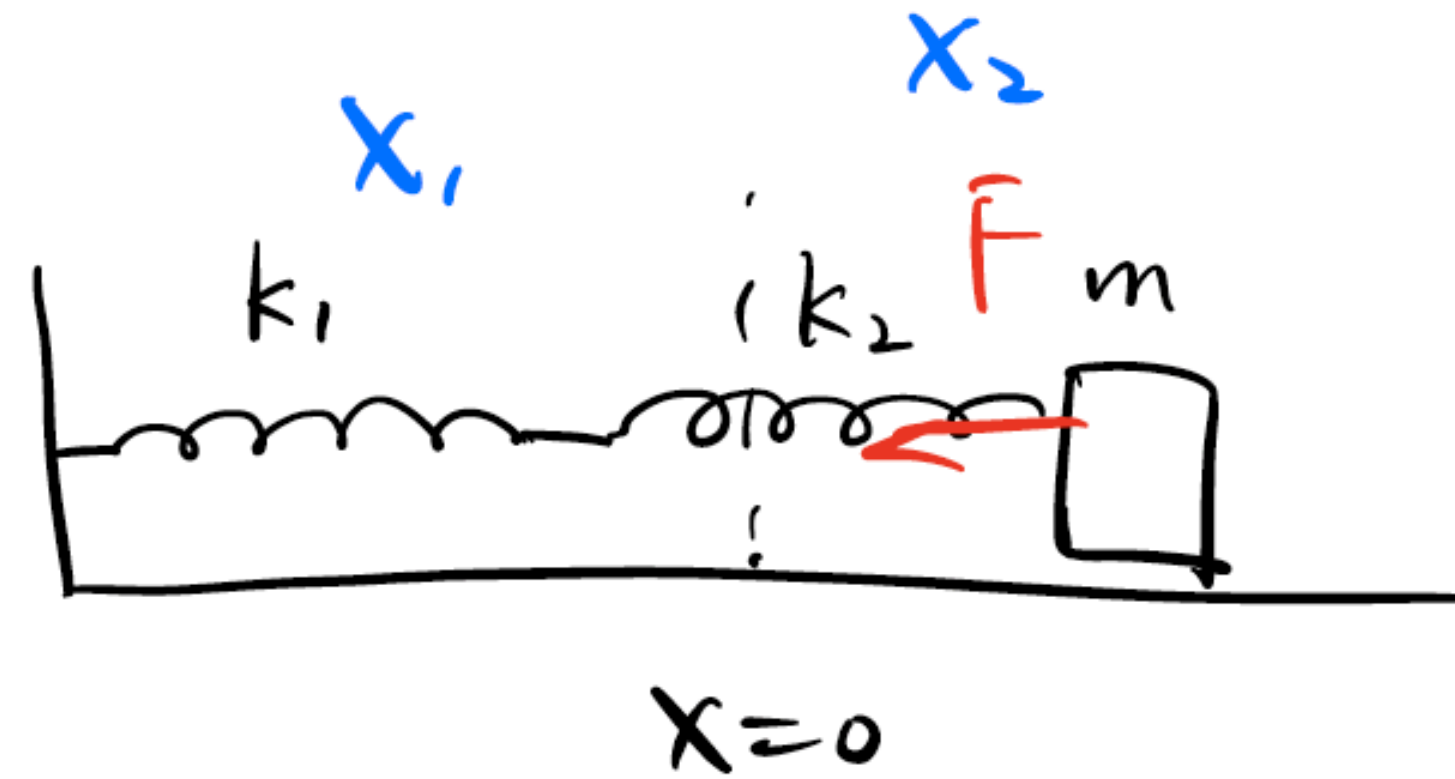
$$k_{\text{eq}} = \sum_i k_i$$

$$\omega = \sqrt{\frac{\sum_i k_i}{m}}$$

$$\omega = \sqrt{\frac{k_1 + k_2 + k_3}{m}}$$

$$F = - (k_1 + k_2 + k_3) x$$

Springs in Series



$$x = x_1 + x_2$$

$$|F| = k_1 x_1 = k_2 x_2$$

$$x_1 = \frac{1}{k_1} \quad x_2 = \frac{1}{k_2}$$

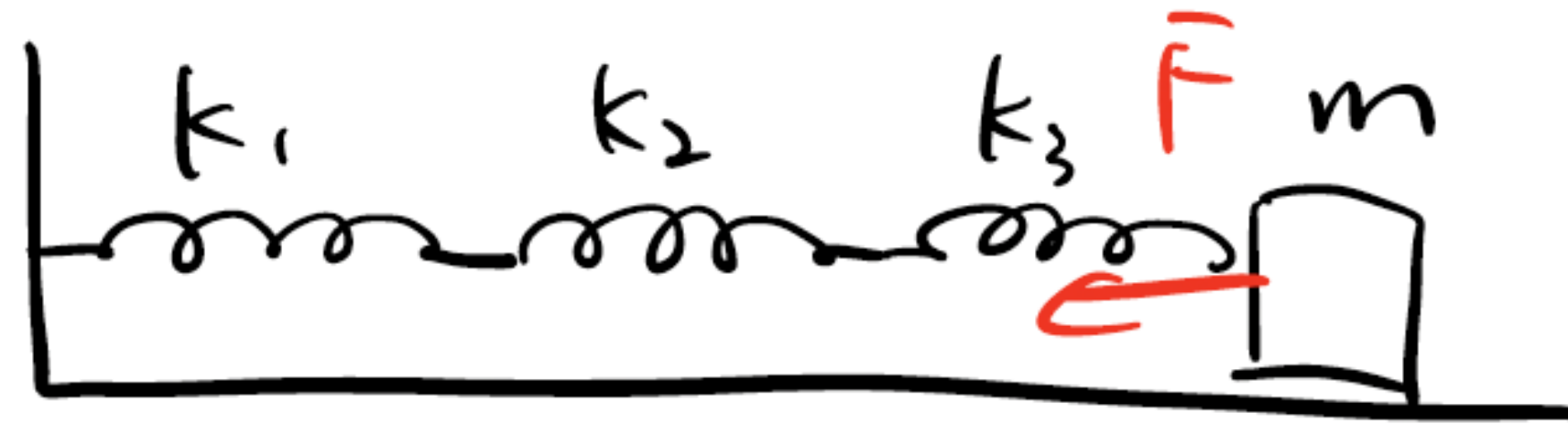
$$x_1 = \frac{\frac{1}{k_1}}{\frac{1}{k_1} + \frac{1}{k_2}} x$$

$$F = - \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}} x = - k_{\text{eff}} x$$

$k_{\text{eff}} = \frac{k_1 k_2}{k_1 + k_2}$

$$\omega = \sqrt{\frac{k_1 k_2}{m(k_1 + k_2)}}$$

Springs in Series

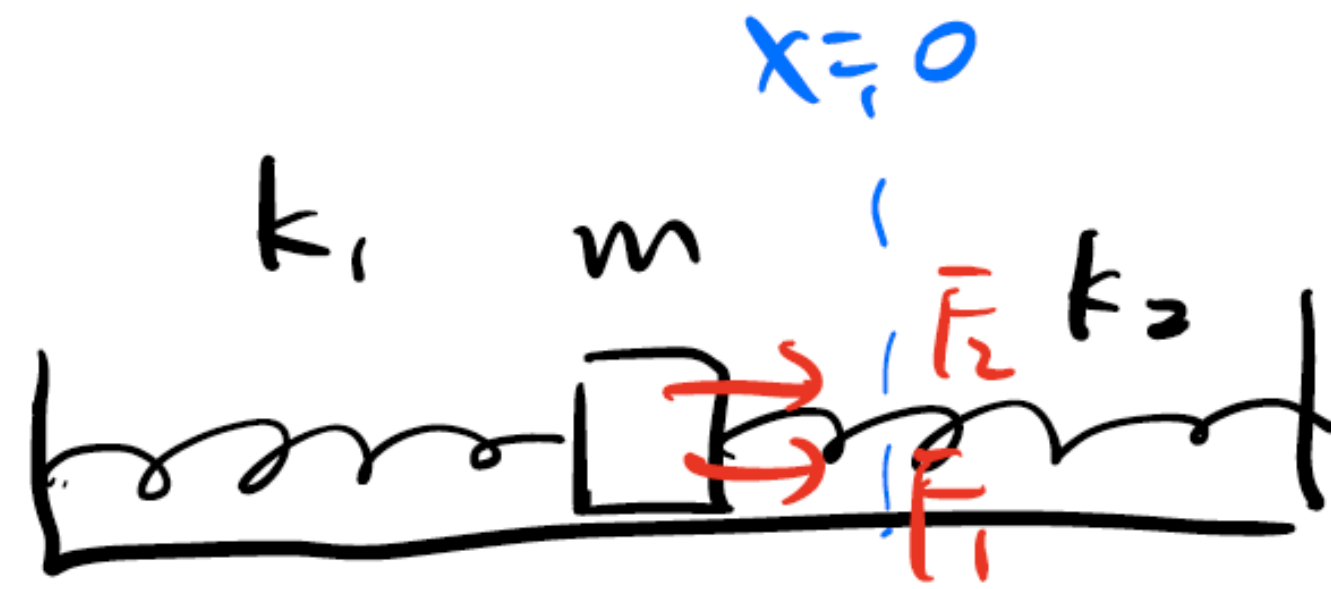


n springs

$$k_{\text{eq}} = \frac{1}{\sum_i k_i}$$

$$\omega = \sqrt{\frac{k_1 k_2 k_3}{k_1 + k_2 + k_3}}$$

Springs of Two Sides



Damped Oscillation

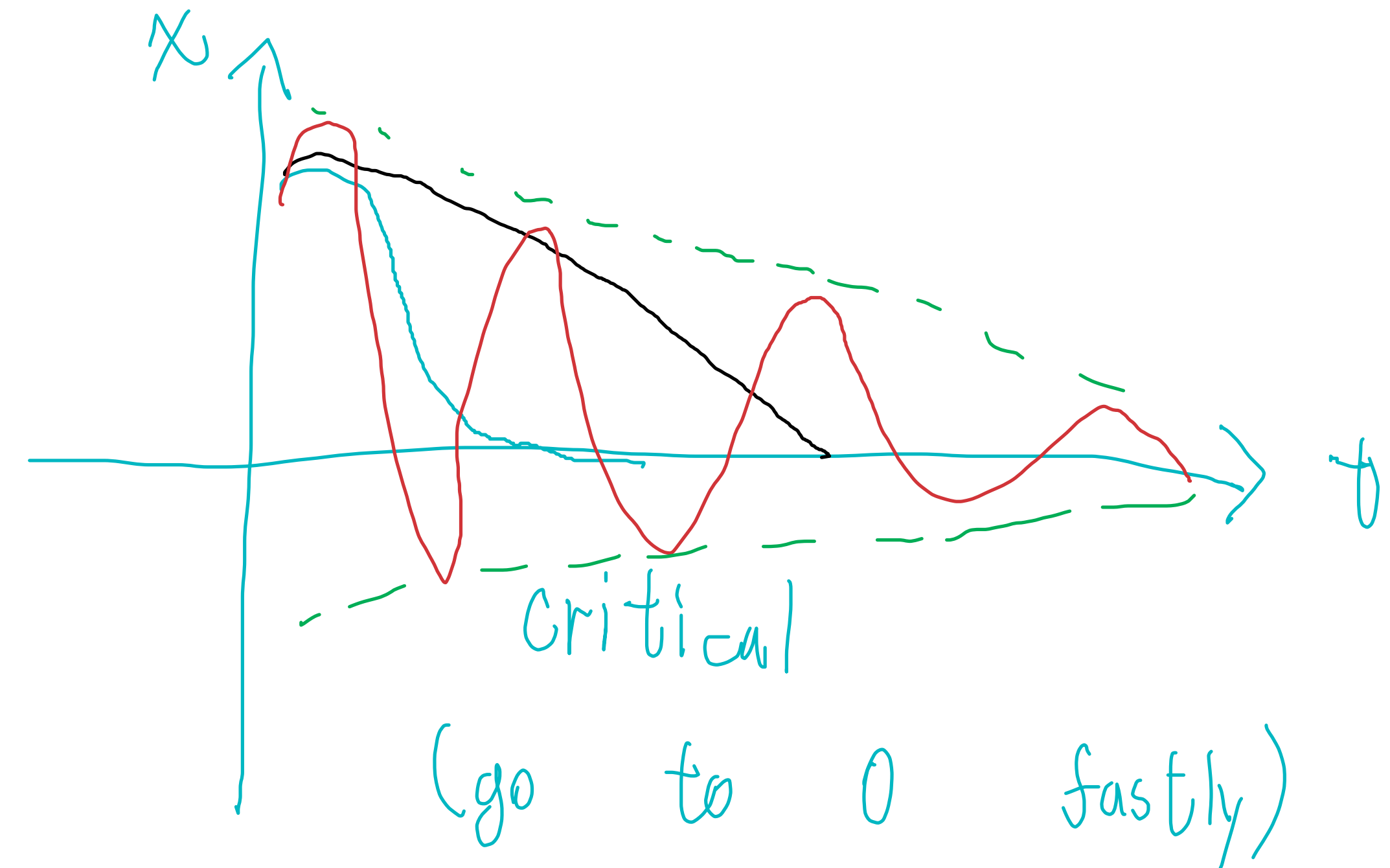
$$\gamma = \frac{b}{m}$$

$$F = ma$$

$$-kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2}$$

Damped Oscillation

- $w_0 < \frac{\gamma}{2}$, overdamped
- $w_0 = \frac{\gamma}{2}$, critical damp
- $w_0 > \frac{\gamma}{2}$, underdamped



overdamped (more time)
underdamped

Q Factor

$$Q = \frac{\omega_0}{\gamma}$$

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$$T = \frac{2\pi}{\omega} \approx \frac{2\pi}{\omega_0}$$

$$A = A_0 e^{-\frac{\gamma}{2} t}$$

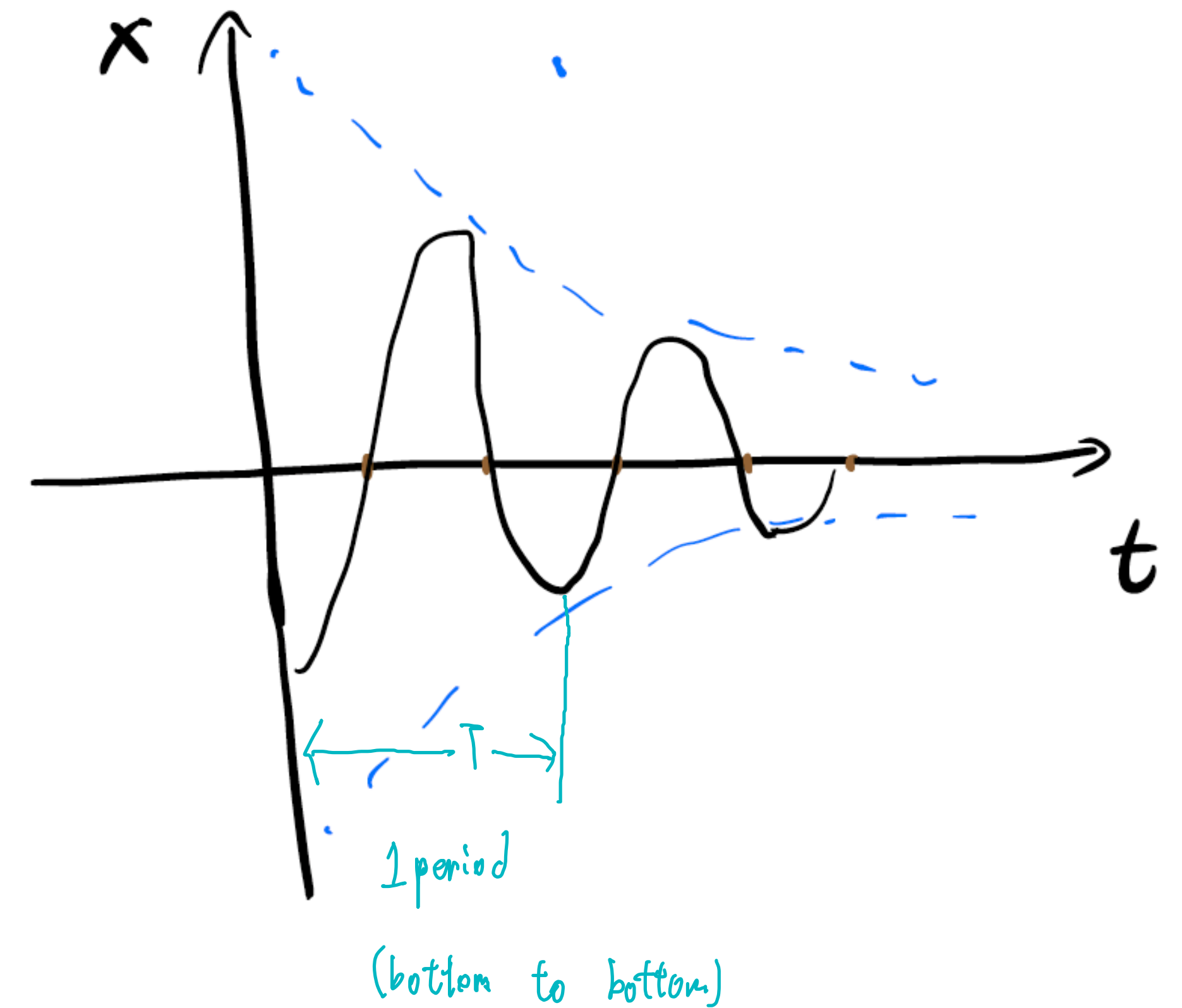
$$A(T) = A_0 e^{-\frac{\gamma}{2} \frac{2\pi}{\omega_0}}$$

$$\frac{E(T)}{E(0)} = \frac{A^2(T)}{A^2(0)} = e^{-\gamma \frac{2\pi}{\omega_0}} = e^{-\frac{2\pi}{Q}}$$

$$\approx 1 - \frac{2\pi}{Q}$$

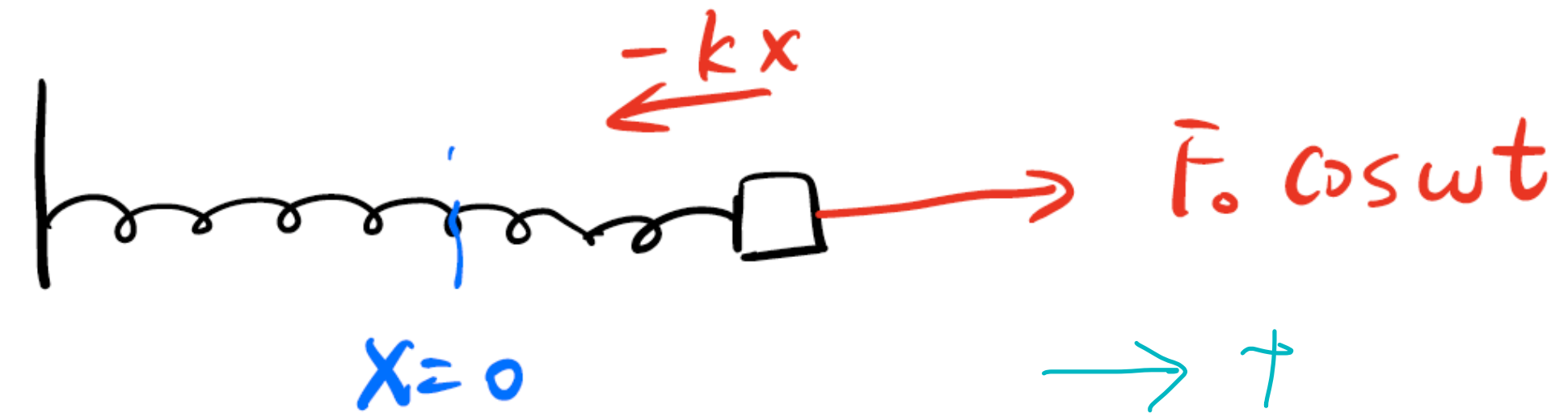
$$\frac{\Delta E}{E} = \frac{2\pi}{Q}$$

$$(e^x \approx 1+x \text{ when } x \rightarrow 0)$$



Forced Oscillation

- $A = \frac{F_0/m}{\omega_0^2 - \omega^2}$
 - $\omega \ll \omega_0$
 - $\omega \gg \omega_0$ ($\omega \rightarrow \infty$)
 - $\omega = \omega_0$, resonance



$$F_0 \cos(\omega t) - kx = m \frac{d^2x}{dt^2}$$

$$-\omega^2 A^2 \cos(\omega t + \varphi) + \omega_0^2 A \cos(\omega t + \varphi) = \frac{F_0}{m} \cos(\omega t)$$

$$\cos(\theta + \varphi)$$

$$= \cos \theta \cos \varphi - \sin \theta \sin \varphi$$

Resonance

- $\omega = \omega_0$, resonance

$\omega \gg \omega_0$: motion and force opposite

$$\phi = \pi$$

$$\cos \phi = -1$$

$$A = 0$$

